

Figure 6.2-19 Feedforward compensation around a noninverting amplifier.

that has a zero in the left half-plane because the gain of the buffer is inverted. Figure 6.2-18(c) can be used as a model for this circuit. The voltage-transfer function $V_{out}(s)/V_{in}(s)$ can be found to be

$$\frac{V_{out}(s)}{V(s)} = \frac{-AC_c}{C_c + C_{II}} \left(\frac{s + g_{mII}/AC_c}{s + 1/[R_{II}(C_c + C_{II})]} \right) \quad (6.2-56)$$

In order to use the circuit in Fig. 6.2-18(b) to achieve compensation, it is necessary to place the zero located at g_{mII}/AC_c above the value of GB so that the boosting of the magnitude will not negate the desired effect of positive phase shift caused by the zero. Fortunately, the phase effects extend over a much broader frequency range than the magnitude effects so that this method will contribute additional phase margin to that provided by the feedback compensation technique. It is quite possible that several zeros can be generated in the transfer function of the op amp. These zeros should all be placed above GB and should be well controlled to avoid large settling times caused by poles and zeros that are close together in the transient response [6].

Another form of feedforward compensation is to provide a feedforward path around a noninverting amplifier. This is shown in Fig. 6.2-19. This type of compensation is often used in source followers, where a capacitor is connected from the gate to source of the source follower. The capacitor will provide a path that bypasses the transistor at high frequencies.

6.3 Design of the Two-Stage Op Amp

The previous two sections described the general approach to op amp design and compensation. In this section, a procedure will be developed that will enable a first-cut design of the two-stage op amp shown in Fig. 6.3-1. In order to simplify the notation, it is convenient to define the notation $S_i = W_i/L_i = (W/L)_i$, where S_i is the ratio of W and L of the i th transistor.

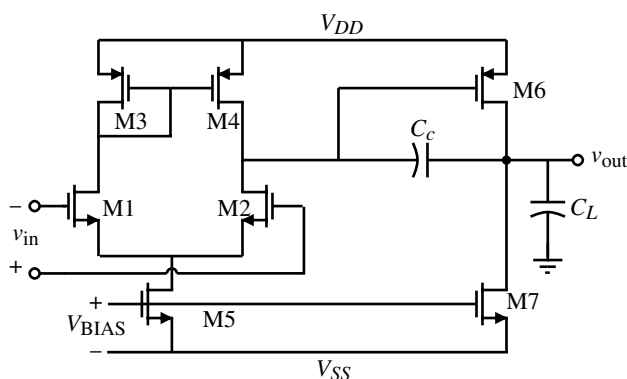


Figure 6.3-1 Schematic of an unbuffered, two-stage CMOS op amp with an n-channel input pair.

Design Procedure for the Two-Stage CMOS Op Amp

Before beginning this task, important relationships describing op amp performance will be summarized from Section 6.2, assuming that $g_{m1} = g_{m2} = g_{m1}$, $g_{m6} = g_{m1}$, $g_{ds2} + g_{ds4} = G_I$, and $g_{ds6} + g_{ds7} = G_{II}$. These relationships are based on the circuit shown in Fig. 6.3-1.

$$\text{Slew rate } SR = \frac{I_5}{C_c} \quad (6.3-1)$$

$$\text{First-stage gain } A_{v1} = \frac{-g_{m1}}{g_{ds2} + g_{ds4}} = \frac{-2g_{m1}}{I_5(\lambda_2 + \lambda_4)} \quad (6.3-2)$$

$$\text{Second-stage gain } A_{v2} = \frac{-g_{m6}}{g_{ds6} + g_{ds7}} = \frac{-g_{m6}}{I_6(\lambda_6 + \lambda_7)} \quad (6.3-3)$$

$$\text{Gain bandwidth } GB = \frac{g_{m1}}{C_c} \quad (6.3-4)$$

$$\text{Output pole } p_2 = \frac{-g_{m6}}{C_L} \quad (6.3-5)$$

$$\text{RHP zero } z_1 = \frac{g_{m6}}{C_c} \quad (6.3-6)$$

$$\text{Positive CMR } V_{in(max)} = V_{DD} - \sqrt{\frac{I_5}{\beta_3}} - |V_{T03}|(max) + V_{T1}(min) \quad (6.3-7)$$

$$\text{Negative CMR } V_{in(min)} = V_{SS} + \sqrt{\frac{I_5}{\beta_1}} + V_{T1}(max) + V_{DS5}(sat) \quad (6.3-8)$$

$$\text{Saturation voltage } V_{DS}(sat) = \sqrt{\frac{2I_{DS}}{\beta}} \quad (6.3-9)$$

It is assumed that all transistors are in saturation for the above relationships.

The following design procedure assumes that specifications for the following parameters are given:

1. Gain at dc, $A_v(0)$
2. Gain bandwidth, GB
3. Input common-mode range, ICMR
4. Load capacitance, C_L
5. Slew rate, SR
6. Output voltage swing
7. Power dissipation, P_{diss}

The design procedure begins by choosing a device length to be used throughout the circuit. This value will determine the value of the channel length modulation parameter λ , which

will be a necessary parameter in the calculation of amplifier gain. Because transistor modeling varies strongly with channel length, the selection of a device length to be used in the design (where possible) allows for more accurate simulation models. Having chosen the nominal transistor device length, one next establishes the minimum value for the compensation capacitor C_c . It was shown in Section 6.2 that placing the output pole p_2 2.2 times higher than the GB permitted a 60° phase margin (assuming that the RHP zero z_1 is placed at or beyond ten times GB). It was shown in Eq. (6.2-22) that such pole and zero placements result in the following requirement for the minimum value for C_c :

$$C_c > (2.2/10)C_L \quad (6.3-10)$$

Next, determine the minimum value for the tail current I_5 , based on slew-rate requirements. Using Eq. (6.3-1), the value for I_5 is determined to be

$$I_5 = SR(C_c) \quad (6.3-11)$$

If the slew-rate specification is not given, then one can choose a value based on settling-time requirements. Determine a value that is roughly ten times faster than the settling-time specification, assuming that the output slews approximately one-half of the supply rail. The value of I_5 resulting from this calculation can be changed later if need be. The aspect ratio of M3 can now be determined by using the requirement for positive input common-mode range. The following design equation for $(W/L)_3$ was derived from Eq. (6.3-7):

$$S_3 = (W/L)_3 = \frac{I_5}{(K'_3)[V_{DD} - V_{in(max)} - |V_{T03}|(max) + V_{T1(min)}]^2} \quad (6.3-12)$$

If the value determined for $(W/L)_3$ is less than one, then it should be increased to a value that minimizes the product of W and L . This minimizes the area of the gate region, which in turn reduces the gate capacitance. This gate capacitance contributes to the mirror pole, which may cause a degradation in phase margin.

Requirements for the transconductance of the input transistors can be determined from knowledge of C_c and GB . The transconductance g_{m1} can be calculated using the following equation:

$$g_{m1} = GB(C_c) \quad (6.3-13)$$

The aspect ratio $(W/L)_1$ is directly obtainable from g_{m1} as shown below:

$$S_1 = (W/L)_1 = \frac{g_{m1}^2}{(K'_1)(I_5)} \quad (6.3-14)$$

Enough information is now available to calculate the saturation voltage of transistor M5. Using the negative ICMR equation, calculate V_{DSS} using the following relationship derived from Eq. (6.3-8):

$$V_{DSS} = V_{in(min)} - V_{SS} - \left(\frac{I_5}{\beta_1}\right)^{1/2} - V_{T1(max)} \quad (6.3-15)$$

If the value for V_{DS5} is less than about 100 mV, then the possibility of a rather large $(W/L)_5$ may result. This may not be acceptable. If the value for V_{DS5} is less than zero, then the ICMR specification may be too stringent. To solve this problem, I_5 can be reduced or $(W/L)_1$ increased. The effects of these changes must be accounted for in previous design steps. One must iterate until the desired result is achieved. With V_{DS5} determined, $(W/L)_5$ can be extracted using Eq. (6.3-9) in the following way:

$$S_5 = (W/L)_5 = \frac{2(I_5)}{K'_5(V_{DS5})^2} \quad (6.3-16)$$

At this point, the design of the first stage of the op amp is complete. We next consider the output stage.

For a phase margin of 60° , the location of the output pole was assumed to be placed at 2.2 times GB . Based on this assumption and the relationship for $|p_2|$ in Eq. (6.3-5), the transconductance g_{m6} can be determined using the following relationship:

$$g_{m6} = 2.2(g_{m2})(C_L/C_c) \quad (6.3-17)$$

Generally, for reasonable phase margin, the value of g_{m6} is approximately ten times the input stage transconductance g_{m1} . At this point, there are two possible approaches to completing the design of M6 (i.e., W_6/L_6 and I_6). The first is to achieve proper mirroring of the first-stage current-mirror load of Fig. 6.3-1 (M3 and M4). This requires that $V_{SG4} = V_{SG6}$. Using the formula for g_m , which is $K'S(V_{GS} - V_T)$, we can write that if $V_{SG4} = V_{SG6}$, then

$$S_6 = S_4 \frac{g_{m6}}{g_{m4}} \quad (6.3-18)$$

Knowing g_{m6} and S_6 will define the dc current I_6 using the following equation:

$$I_6 = \frac{g_{m6}^2}{(2)(K'_6)(W/L)_6} = \frac{g_{m6}^2}{2K'_6 S_6} \quad (6.3-19)$$

One must now check to make sure that the maximum output voltage specification is satisfied. If this is not true, then the current or W/L ratio can be increased to achieve a smaller $V_{DS}(\text{sat})$. If these changes are made to satisfy the maximum output voltage specification, then the proper current mirroring of M3 and M4 is no longer guaranteed.

The second approach to designing the output stage is to use the value of g_{m6} and the required $V_{DS}(\text{sat})$ of M6 to find the current. Combining the defining equation for g_m and $V_{DS}(\text{sat})$ results in an equation relating (W/L) , $V_{DS}(\text{sat})$, g_m , and process parameters. Using this relationship, given below, with the $V_{DS}(\text{sat})$ requirement taken from the output range specification one can determine $(W/L)_6$.

$$S_6 = (W/L)_6 = \frac{g_{m6}}{K'_6 V_{DS6}(\text{sat})} \quad (6.3-20)$$

Equation (6.3-19) is used as before to determine a value for I_6 . In either approach to finding I_6 , one also should check the power dissipation requirements since I_6 will most likely determine the majority of the power dissipation.

The device size of M7 can be determined from the balance equation given below:

$$S_7 = (W/L)_7 = (W/L)_5 \left(\frac{I_6}{I_5} \right) = S_5 \left(\frac{I_6}{I_5} \right) \quad (6.3-21)$$

The first-cut design of all W/L ratios is now complete. Figure 6.3-2 illustrates the above design procedure showing the various design relationships and where they apply in the two-stage CMOS op amp.

At this point in the design procedure, the total amplifier gain must be checked against the specifications.

$$A_v = \frac{(2)(g_{m2})(g_{m6})}{I_5(\lambda_2 + \lambda_4)I_6(\lambda_6 + \lambda_7)} \quad (6.3-22)$$

If the gain is too low, a number of things can be adjusted. The best way to do this is to use Table 6.3-1, which shows the effects of various device sizes and currents on the different parameters generally specified. Each adjustment may require another pass through this design procedure in order to ensure that all specifications have been met. Table 6.3-2 summarizes the above design procedure.

No attempt has been made to account for noise or PSRR thus far in the design procedure. Now that the preliminary design is complete, these two specifications can be addressed. The input-referred noise voltage results primarily from the load and input transistors of the first stage. Each of these contributes both thermal and $1/f$ noise. The $1/f$ noise contributed by any transistor can be reduced by increasing device area (e.g., increase WL). Thermal noise contributed by any transistor can be reduced by increasing its g_m . This is accomplished by an increase in W/L , an increase in current, or both. Effective input-noise voltage attributed to the load transistors can be reduced by reducing the g_{m3}/g_{m1} (g_{m4}/g_{m2}) ratio. One must be careful that these adjustments to improve noise performance do not adversely affect some other important performance parameter of the op amp.

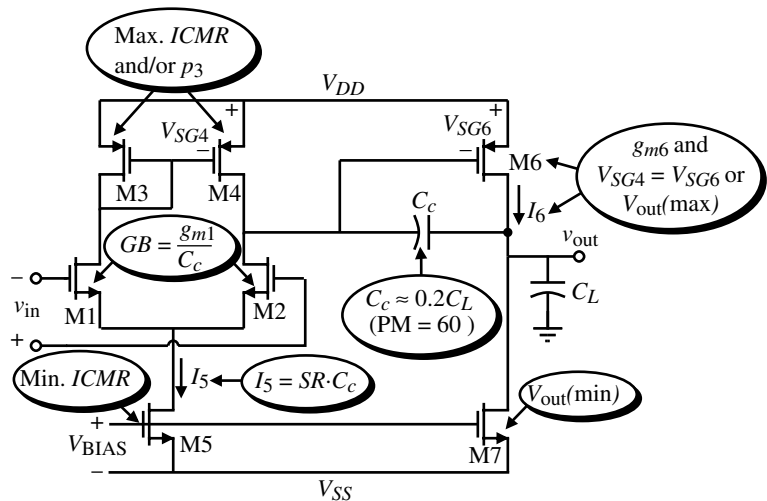


Figure 6.3-2 Illustration of the design relationship and the circuit for a two-stage CMOS op amp.

Table 6.3-1 Dependence of the Performance of Fig. 6.3-1 on dc Current, W/L Ratios, and the Compensating Capacitor

	Drain Current		M1 and M2		M3 and M4		Inverter	Inverter Load		Compensation Capacitor
	I_5	I_7	W/L	L	W	L	W_6/L_6	W_7	L_7	C_c
Increase dc Gain	$(\downarrow)^{1/2}$	$(\downarrow)^{1/2}$	$(\uparrow)^{1/2}$	$\uparrow$		$\uparrow$	$(\uparrow)^{1/2}$		$\uparrow$	
Increase GB	$(\uparrow)^{1/2}$		$(\uparrow)^{1/2}$							$\downarrow$
Increase RHP Zero		$(\uparrow)^{1/2}$					$(\uparrow)^{1/2}$			$\downarrow$
Increase Slew Rate	$\uparrow$									$\downarrow$
Increase C_L										$\uparrow$

Table 6.3-2 Unbuffered Op Amp Design Procedure

This design procedure assumes that the gain at dc (A_v), unity-gain bandwidth (GB), input common-mode range [$V_{in}(\min)$ and $V_{in}(\max)$], load capacitance (C_L), slew rate (SR), settling time (T_s), output voltage swing [$V_{out}(\max)$ and $V_{out}(\min)$], and power dissipation (P_{diss}) are given.

1. Choose the smallest device length that will keep the channel modulation parameter constant and give good matching for current mirrors.
2. From the desired phase margin, choose the minimum value for C_c ; that is, for a 60° phase margin we use the following relationship. This assumes that $z \geq 10 GB$.

$$C_c > 0.22C_L$$

3. Determine the minimum value for the “tail current” (I_5) from the largest of the two values.

$$I_5 = SR \cdot C_c$$

4. Design for S_3 from the maximum input voltage specification.

$$S_3 = \frac{2I_3}{K'_3[V_{DD} - V_{in}(\max) - |V_{T03}|(\max) + V_{T1}(\min)]^2} \geq 1$$

5. Verify that the pole and zero due to C_{gs3} and C_{gs4} ($= 0.67W_3L_3C_{ox}$) will not be dominant by assuming p_3 to be greater than $10GB$. Note that a zero exists at $2p_3$ and will diminish the influence of p_3 on the op amp.

$$\frac{g_{m3}}{2C_{gs3}} > 10 GB$$

6. Design for S_1 (S_2) to achieve the desired GB .

$$g_{m1} = GB \cdot C_c \Rightarrow S_1 = S_2 = \frac{g_{m2}^2}{K'_2 I_5}$$

Table 6.3-2 (continued)

7. Design for S_5 from the minimum input voltage. First calculate $V_{DS5}(\text{sat})$ and then find S_5 .

$$V_{DS5}(\text{sat}) = V_{in}(\text{min}) - V_{SS} - \sqrt{\frac{I_5}{\beta_1}} - V_{T1}(\text{max}) \geq 100 \text{ mV}$$

$$S_5 = \frac{2I_5}{K'_5[V_{DS5}(\text{sat})]^2}$$

8. Find S_6 and I_6 by letting the second pole (p_2) be equal to 2.2 times GB .

$$g_{m6} = 2.2g_{m2}(C_L/C_c)$$

Let $V_{SG4} = V_{SG6}$, which gives

$$S_6 = S_4 \frac{g_{m6}}{g_{m4}}$$

Knowing g_{m6} and S_6 allows us to solve for I_6 as

$$I_6 = \frac{g_{m6}^2}{2K'_6 S_6}$$

9. Alternately, I_6 can be calculated by solving for S_6 using

$$S_6 = \frac{g_{m6}}{K'_6 V_{DS6}(\text{sat})}$$

and then using the previous relationship to find I_6 . Of course, the proper mirror between M3 and M4 is no longer guaranteed.

10. Design S_7 to achieve the desired current ratios between I_5 and I_6 .

$$S_7 = (I_6/I_5)S_5$$

11. Check gain and power dissipation specifications.

$$A_v = \frac{2g_{m2}g_{m6}}{I_5(\lambda_2 + \lambda_3)(\lambda_6 + \lambda_7)}$$

$$P_{\text{diss}} = (I_5 + I_6)(V_{DD} + |V_{SS}|)$$

12. If the gain specification is not met, then the currents I_5 and I_6 can be decreased or the W/L ratios of M2 and/or M6 increased. The previous calculations must be rechecked to ensure that they have been satisfied. If the power dissipation is too high, then one can only reduce the currents I_5 and I_6 . Reduction of currents will probably necessitate an increase of some of the W/L ratios to satisfy input and output swings.

13. Simulate the circuit to check to see that all specifications are met.

The power-supply rejection ratio is to a large degree determined by the configuration used. Some improvement in negative PSRR can be achieved by increasing the output resistance of M5. This is usually accomplished by increasing both W_5 and L_5 proportionately without seriously affecting any other performance. Transistor M7 should be adjusted accordingly for proper matching. A more detailed analysis of the PSRR of the two-stage op amp will be considered in the next section.

The following example illustrates the steps in designing the op amp described.

Example 6.3-1

Design of a Two-Stage Op Amp

Using the material and device parameters given in Tables 3.1-1 and 3.1-2, design an amplifier similar to that shown in Fig. 6.3-1 that meets the following specifications with a phase margin of 60° . Assume the channel length is to be $1\text{ }\mu\text{m}$.

$$\begin{array}{lll} A_v > 5000\text{ V/V} & V_{DD} = 2.5\text{ V} & V_{SS} = -2.5\text{ V} \\ GB = 5\text{ MHz} & C_L = 10\text{ pF} & SR > 10\text{ V}/\mu\text{s} \\ V_{\text{out range}} = \pm 2\text{ V} & \text{ICMR} = -1\text{ to }2\text{ V} & P_{\text{diss}} \leq 2\text{ mW} \end{array}$$

SOLUTION

The first step is to calculate the minimum value of the compensation capacitor C_c , which is

$$C_c > (2.2/10)(10\text{ pF}) = 2.2\text{ pF}$$

Choose C_c as 3 pF. Using the slew-rate specification and C_c calculate I_5 .

$$I_5 = (3 \times 10^{-12})(10 \times 10^6) = 30\text{ }\mu\text{A}$$

Next calculate $(W/L)_3$ using ICMR requirements. Using Eq. (6.3-12) we have

$$(W/L)_3 = \frac{30 \times 10^{-6}}{(50 \times 10^{-6})[2.5 - 2 - 0.85 + 0.55]^2} = 15$$

Therefore,

$$(W/L)_3 = (W/L)_4 = 15$$

Now we can check the value of the mirror pole, p_3 , to make sure that it is in fact greater than $10GB$. Assume the $C_{\text{ox}} = 2.47\text{ fF}/\mu\text{m}^2$. The mirror pole can be found as

$$p_3 \approx \frac{-g_{m3}}{2C_{gs3}} = \frac{-\sqrt{2K_p S_3 I_3}}{2(0.667) W_3 L_3 C_{\text{ox}}} = 2.81 \times 10^9\text{ rad/s}$$

or 448 MHz. Thus, p_3 and z_3 are not of concern in this design because $p_3 \gg 10GB$.

The next step in the design is to calculate g_{m1} using Eq. (6.3-13).

$$g_{m1} = (5 \times 10^6)(2\pi)(3 \times 10^{-12}) = 94.25 \mu\text{S}$$

Therefore, $(W/L)_1$ is

$$(W/L)_1 = (W/L)_2 = \frac{g_{m1}^2}{2K'_N I_1} = \frac{(94.25)^2}{2 \cdot 110 \cdot 15} = 2.79 \approx 3.0$$

Next, calculate V_{DS5} using Eq. (6.3-15).

$$V_{DS5} = (-1) - (-2.5) - \sqrt{\frac{30 \times 10^{-6}}{110 \times 10^{-6} \cdot 3}} - 0.85 = 0.35 \text{ V}$$

Using V_{DS5} calculate $(W/L)_5$ from Eq. (6.3-16).

$$(W/L)_5 = \frac{2(30 \times 10^{-6})}{(110 \times 10^{-6})(0.35)^2} = 4.49 \approx 4.5$$

From Eq. (6.2-20), we know that

$$g_{m6} \geq 10g_{m1} \geq 942.5 \mu\text{S}$$

Assuming that $g_{m6} = 942.5 \mu\text{S}$ and calculating g_{m4} as $150 \mu\text{S}$, we use Eq. (6.3-18) to get

$$S_6 = S_4 \frac{g_{m6}}{g_{m4}} = 15 \cdot \frac{942.5}{150} = 94.25 \approx 94$$

Calculate I_6 using Eq. (6.3-19).

$$I_6 = \frac{(942.5 \times 10^{-6})^2}{(2)(50 \times 10^{-6})(94)} = 94.5 \mu\text{A} \approx 95 \mu\text{A}$$

Designing S_6 by using Eq. (6.3-20) gives $S_6 \approx 15$. Since the W/L ratio of 94 from above is greater, the maximum output voltage specification will be met.

Finally, calculate $(W/L)_7$ using Eq. (6.3-21).

$$(W/L)_7 = 4.5 \left(\frac{95 \times 10^{-6}}{30 \times 10^{-6}} \right) = 14.25 \approx 14$$

Let us check the $V_{\text{out(min)}}$ specification although the W/L of M7 is large enough that this is probably not necessary. The value of $V_{\text{out(min)}}$ is

$$V_{\text{min(out)}} = V_{DS7(\text{sat})} = \sqrt{\frac{2 \cdot 95}{110 \cdot 14}} = 0.351 \text{ V}$$

which is less than required. At this point, the first-cut design is complete.

The power dissipation can be calculated as

$$P_{\text{diss}} = 5 \text{ V} \cdot (30 \mu\text{A} + 95 \mu\text{A}) = 0.625 \text{ mW}$$

Now check to see that the gain specification has been met.

$$A_v = \frac{(2)(92.45 \times 10^{-6})(942.5 \times 10^{-6})}{30 \times 10^{-6}(0.04 + 0.05)95 \times 10^{-6}(0.04 + 0.05)} = 7696 \text{ V/V}$$

which meets specifications. If more gain were desired, an easy way to achieve it would be to increase the W and L values by a factor of 2, which because of the decreased value of λ would increase the gain by a factor of 20. Figure 6.3-3 shows the results of the first-cut design. The next phase requires simulation.

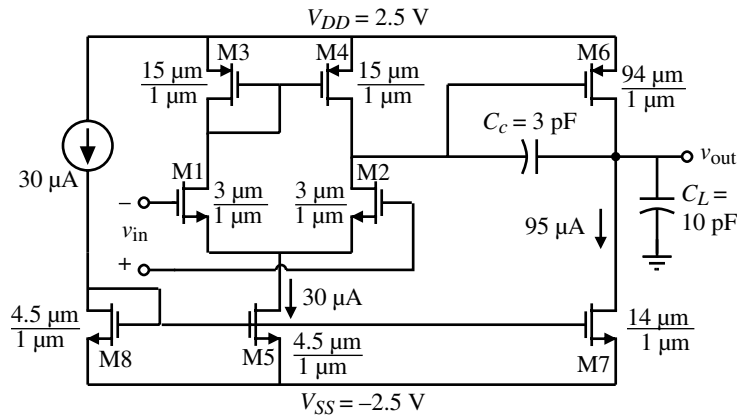


Figure 6.3-3 Result of Example 6.3-1.

Nulling Resistor, Miller Compensation

It may likely occur that the undesired RHP zero may not be negligible in the above design procedure. This would occur if the GB specification was large or if the output stage transconductance (g_{m6}) was not large. In this case, it becomes necessary to employ the nulling resistor compensation method. We shall use the results of Section 6.2 to illustrate how to apply this solution.

Section 6.2 described a technique whereby the RHP zero can be moved to the left half-plane and placed on the highest nondominant pole. To accomplish this, a resistor is placed in series with the compensation capacitor. Figure 6.3-4 shows a compensation scheme using transistor M8 as a resistor. This transistor is controlled by a control voltage V_c that adjusts the resistor so that it maintains the proper value over process variations [6].

With the addition of the resistor in the compensation scheme, the resulting poles and zeros are [see Eqs. (6.2-37) to (6.2-40)]

$$p_1 = -\frac{g_{m2}}{A_v C_c} = -\frac{g_{m1}}{A_v C_c} \quad (6.3-23)$$

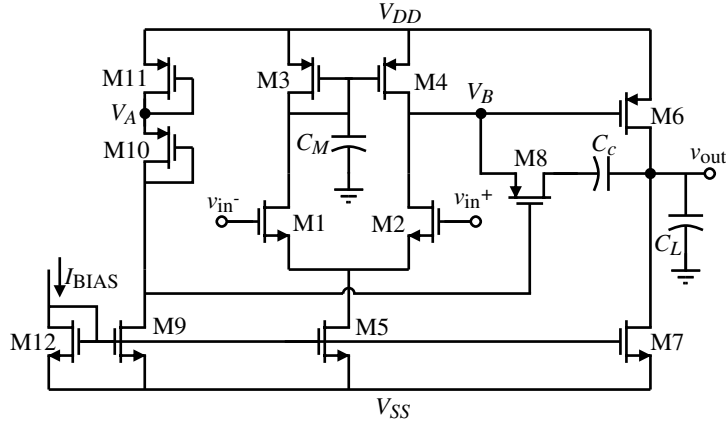


Figure 6.3-4 CMOS two-stage op amp using nulling resistor compensation.

$$p_2 = -\frac{g_{m6}}{C_L} \quad (6.3-24)$$

$$p_4 = -\frac{1}{R_z C_I} \quad (6.3-25)$$

$$z_1 = \frac{-1}{R_z C_c - C_c / g_{m6}} \quad (6.3-26)$$

where $A_v = g_{m1}g_{m6}R_I R_{II}$. In order to place the zero on top of the second pole (p_2), the following relationship must hold:

$$R_z = \frac{1}{g_{m6}} \left(\frac{C_L + C_c}{C_c} \right) = \left(\frac{C_c + C_L}{C_c} \right) \frac{1}{\sqrt{2K'_p S_6 I_6}} \quad (6.3-27)$$

The resistor R_z is realized by the transistor M8, which is operating in the active region because the dc current through it is zero. Therefore, R_z can be written as

$$R_z = \left. \frac{\partial v_{DS8}}{\partial i_{D8}} \right|_{v_{DS8}=0} = \frac{1}{K'_p S_8 (V_{SG8} - |V_{TP}|)} \quad (6.3-28)$$

The bias circuit is designed so that voltage V_A is equal to V_B . As a result,

$$|V_{GS10}| - |V_T| = |V_{GS8}| - |V_T| \quad (6.3-29)$$

In the saturation region

$$|V_{GS10}| - |V_T| = \sqrt{\frac{2(I_{10})}{K'_p (W_{10}/L_{10})}} = |V_{GS8}| - |V_T| \quad (6.3-30)$$

Substituting into Eq. (6.3-28) yields

$$R_z = \frac{1}{K'_p S_8} \sqrt{\frac{K'_p S_{10}}{2I_{10}}} = \frac{1}{S_8} \sqrt{\frac{S_{10}}{2K'_p I_{10}}} \quad (6.3-31)$$

Equating Eq. (6.3-27) to Eq. (6.3-31) gives

$$\left(\frac{W_8}{L_8}\right) = \left(\frac{C_c}{C_L + C_c}\right) \sqrt{\frac{S_{10} S_6 I_6}{I_{10}}} \quad (6.3-32)$$

This relationship must be met in order for Eq. (6.3-27) to hold. To complete the design of this compensation circuit, M11 must be designed to meet the criteria set forth in Eq. (6.3-29). To accomplish this V_{SG11} must be equal to V_{SG6} . Therefore,

$$\left(\frac{W_{11}}{L_{11}}\right) = \left(\frac{I_{10}}{I_6}\right) \left(\frac{W_6}{L_6}\right) \quad (6.3-33)$$

The example that follows illustrates the design of this compensation scheme.

Example 6.3-2

RHP Zero Compensation

Use results of Example 6.3-1 and design compensation circuitry so that the RHP zero is moved from the RHP to the LHP and placed on top of the output pole p_2 . Use device data given in Example 6.3-1.

SOLUTION

The task at hand is the design of transistors M8, M9, M10, M11, and bias current I_{10} . The first step in this design is to establish the bias components. In order to set V_A equal to V_B , V_{SG10} must be equal to V_{SG6} . Therefore,

$$S_{11} = (I_{11}/I_6)S_6$$

Choose $I_{11} = I_{10} = I_9 = 15 \mu\text{A}$, which gives $S_{11} = (15 \mu\text{A}/95 \mu\text{A}) \cdot 94 = 14.8 \approx 15$.

The aspect ratio of M10 is essentially a free parameter and will be set equal to 1. There must be sufficient supply voltage to support the sum of V_{SG11} , V_{SG10} , and V_{DS9} . The ratio of I_{10}/I_5 determines the (W/L) of M9. This ratio is

$$(W/L)_9 = (I_{10}/I_5)(W/L)_5 = (15/30)(4.5) = 2.25 \approx 2$$

Now using design Eq. (6.3-32), $(W/L)_8$ is determined to be

$$(W/L)_8 = \left(\frac{3 \text{ pF}}{3 \text{ pF} + 10 \text{ pF}}\right) \sqrt{\frac{1 \cdot 94 \cdot 95 \mu\text{A}}{15 \mu\text{A}}} = 5.63 \approx 6$$

It is worthwhile to check that the RHP zero has been moved on top of p_2 . To do this, first calculate the value of R_z . V_{SG8} must first be determined. It is equal to V_{SG10} , which is

$$V_{SG10} = \sqrt{\frac{2I_{10}}{K'_p S_{10}}} + |V_{TP}| = \sqrt{\frac{2 \cdot 15}{50 \cdot 1}} + 0.7 = 1.474 \text{ V}$$

Next, determine R_z .

$$R_z = \frac{1}{K'_p S_8 (V_{SG10} - |V_{TP}|)} = \frac{10^6}{50 \cdot 5.63(1.474 - 0.7)} = 4.590 \text{ k}\Omega$$

Using Eq. (6.3-26), the location of z_1 is calculated to be

$$z_1 = \frac{-1}{(4.590 \times 10^3)(3 \times 10^{-12}) - \frac{3 \times 10^{-12}}{942.5 \times 10^{-6}}} = -94.46 \times 10^6 \text{ rad/s}$$

The output pole, p_2 , is found from Eq. (6.3-24) and is

$$p_2 = \frac{942.5 \times 10^{-6}}{10 \times 10^{-12}} = -94.25 \times 10^6 \text{ rad/s}$$

Thus, we see that, for all practical purposes, the output pole is canceled by the zero that has been moved from the RHP to the LHP.

The results of this design are summarized below.

$$W_8 = 6 \text{ }\mu\text{m}$$

$$W_9 = 2 \text{ }\mu\text{m}$$

$$W_{10} = 1 \text{ }\mu\text{m}$$

$$W_{11} = 15 \text{ }\mu\text{m}$$

Because we are trying to cancel a pole with a zero, let us examine the dependence of the nulling resistor technique on temperature and the process variations. The key relationship is given in Eq. (6.3-26), where R_z is canceling $1/g_{m6}$. If we think of the technique as g_{m6} canceling $1/R_z$, the answer is clear. One of the forms for the small-signal transconductance is

$$g_{m6} = K'_p (W_6/L_6) (V_{SG6} - |V_{TP}|) \quad (6.3-34)$$

From Eq. (6.3-28), we see that $1/R_z$ has exactly the same form. As long as we keep M6, M8, and M10 the same type of transistor, the temperature and process tracking should be good. One could replace the resistor by an MOS diode and also achieve good temperature and process tracking (see Problem 6.3-13).

Simulation of the Electrical Design

After the design has been developed using the above procedures, the next step is to simulate the circuit using more accurate models of the transistor. In most cases, the BSIM2 [7] or BSIM3 [8] model is sufficient. The designer should be sure the computer simulation makes sense with the hand design. The computer simulation will take into account many of the details that have been neglected such as the bulk effect. While the designer can make minor modifications in the design, one should resist the temptation to use the computer to make major design changes, such as an architecture change. The function of simulation is to verify and to explore the influence of things like matching, process variations, temperature changes, and power-supply changes. If one does not have information on how the process parameters vary, a good substitute is to examine the circuit at high and low temperatures. These simulations will give some idea of the circuit's dependence on process parameter changes. Also, simulators can be used with Monte Carlo methods to investigate the influence of statistical variations in the values of components.

Although the circuit has not yet been designed physically, it is a good practice to include some of the parasitics that will be due to the physical layout. This will minimize the differences between the simulation performance before layout and after layout. At this point, the designer only knows the W/L values and the dc currents. The area and shape of the source and drain are not yet determined. Unfortunately, the parasitic capacitances due to the reverse-biased bulk–source and bulk–drain cannot be modeled until the area and periphery of the source and drain are known.

A technique that will allow the designer to include the bulk–source and drain–source capacitance parasitics is outlined below. In Fig. 6.3-5, a simple rectangular MOSFET layout is shown. The minimum possible source or drain area would be that indicated by the sum of the lengths $L1$, $L2$, and $L3$ times W . The lengths $L1$, $L2$, and $L3$ are related to the design rules for a given process and are as follows:

$L1$ = Minimum allowable distance between the contact in S/D and the poly

$L2$ = Width of a minimum size contact to diffusion

$L3$ = Minimum allowable distance from the contact in S/D to the edge of the S/D

These rules are easily found from the technology information. Thus, the minimum area of the drain and source is $(L1 + L2 + L3) \times W$ and the corresponding periphery is $2(L1 + L2 + L3) + 2W$. A conservative approach might be to double this area to account for connection parasitics between the source (drain) and their respective destinations.

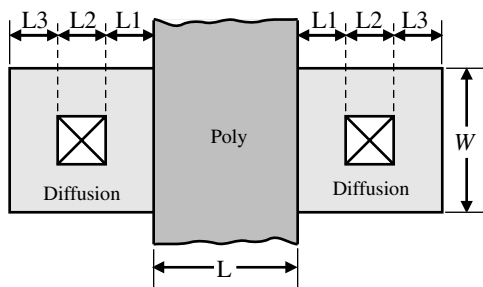


Figure 6.3-5 Method of estimating the source and drain areas and peripheries.